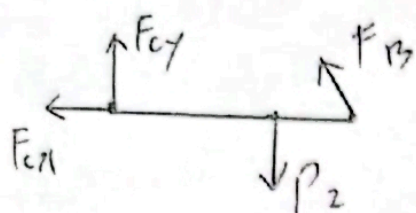
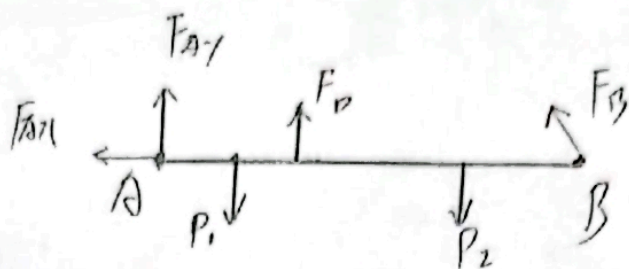


1-2

e)

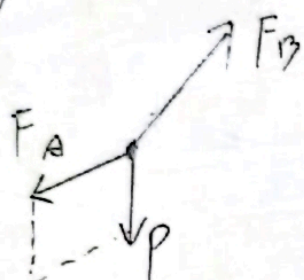


梁 CB

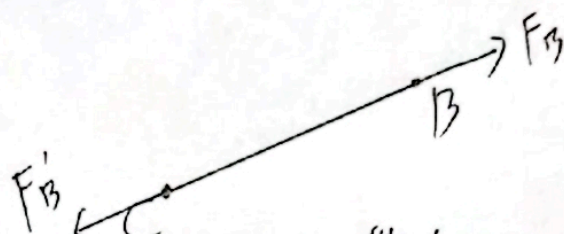


梁 AB

f)

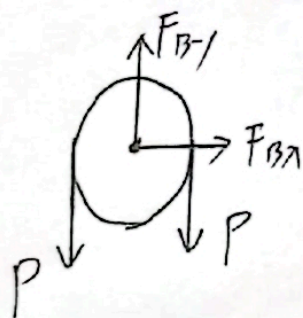


销 B

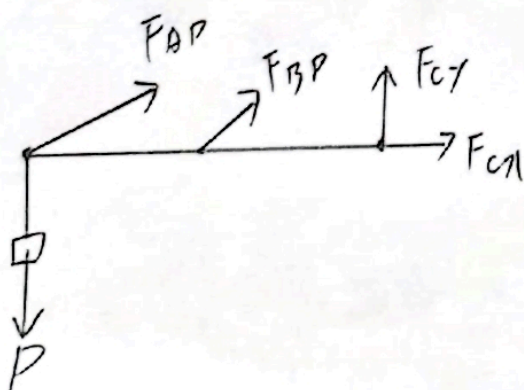


杆 BC (带销)

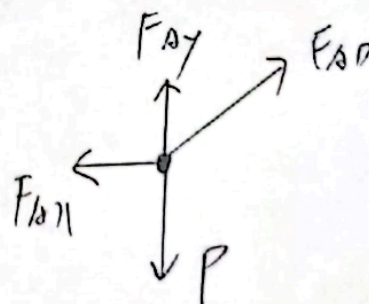
g)



h)



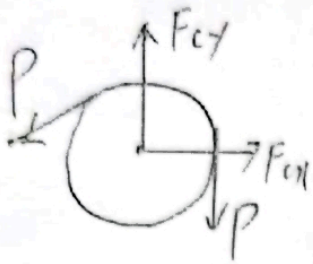
重物连同杆 AC



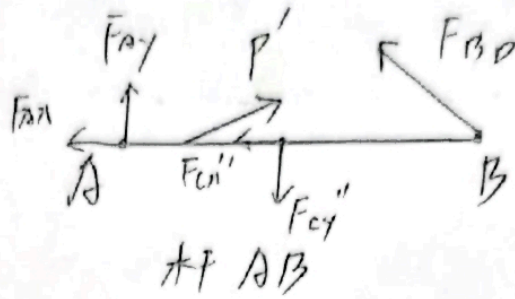
销 A

1-3

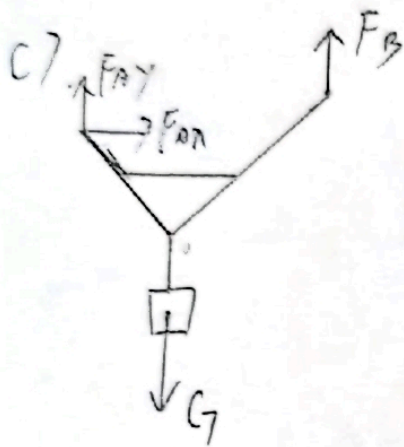
b)



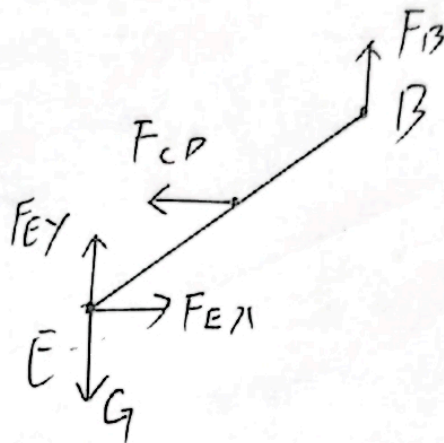
轮 C



杆 AB

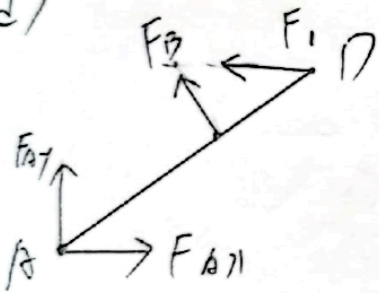


整体

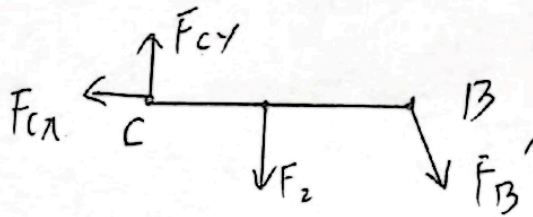


杆 BE (带销 E)

d)

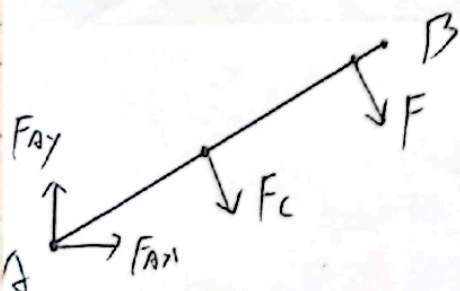


摇杆 AD

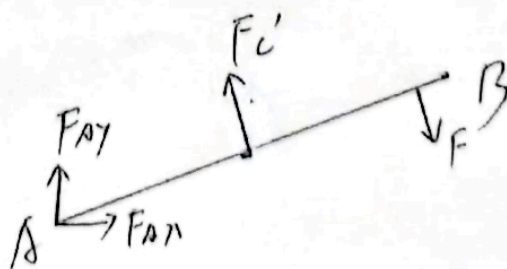


杆 CB 连同滑块

f)



杆 AB



杆 AB 连同套管 C

$$14 \quad a) \quad M(0) = 50 \times 1 \times \sin 60^\circ = 25\sqrt{3} \text{ N}\cdot\text{m} \quad \underline{\text{逆}}$$

$$\sum F_x = \sum F_y = 0$$

$$b) \quad M(0) = 50 \times 1 = 50 \text{ N}\cdot\text{m} \quad \underline{\text{逆}}$$

$$\sum F_x = \sum F_y = 0$$

$$c) \quad M(0) = 50 \times 1 = 50 \text{ N}\cdot\text{m} \quad \underline{\text{顺}}$$

$$\sum F_x = \sum F_y = 0$$

1-8 将力在 xOy 平面分解 F_x 和 F_y

$$\begin{cases} M_x(F) = F_y \cdot 100 = -36 \text{ N}\cdot\text{m} \end{cases}$$

$$\begin{cases} M_z(F) = F_x \cdot (100 + 50) + F_y \cdot 70 = 20.2 \text{ N}\cdot\text{m} \end{cases}$$

$$\therefore \begin{cases} F_x = 0.3 \text{ N} \\ F_y = 0.36 \text{ N} \end{cases}$$

$$\therefore M_y(F) = F_x \cdot 100 = 30 \text{ N}\cdot\text{m}$$

3-4 A处为固定端约束, 可化为矢E M_A 和 F_{Ax} , F_{Ay}

$$\therefore \begin{cases} F_{Ax} = 0 \\ F_{Ay} = 3 \times 4 + 5 = 17 \text{ kN} \\ M_A = 10 + 3 \times 4 \times \frac{3}{2} + 5 \times 3 = 43 \text{ kN}\cdot\text{m} \end{cases}$$

3-5

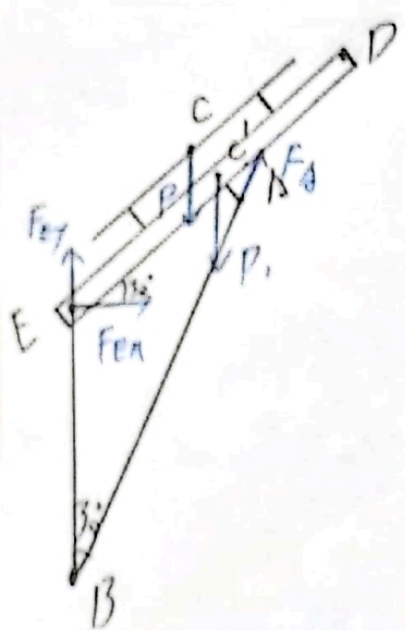
A处受 F_{Ax} , F_{Ay} , 绳上受力为 P , 杆BC受力为 F_{BC}

$$\therefore \begin{cases} F_{Ay} + F_{BC} \cdot \cos 45^\circ = P \\ F_{Ax} = P + F_{BC} \cdot \sin 45^\circ \\ \sum M_B(F) = 0: P \times 0.1 + F_{BC} \cdot \sin 45^\circ \times 0.6 = P \times 0.3 \end{cases}$$

$$\therefore \begin{cases} F_{Ax} = 1200 \text{ N} \\ F_{Ay} = 2400 \text{ N} \\ F_{BC} = 600\sqrt{2} \text{ N} \end{cases}$$

3-9

受力分析如图:



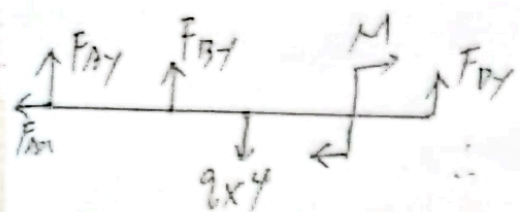
$$\therefore \begin{cases} F_{Ey} + F_D \cdot \cos 30^\circ = P + P_1 \\ F_{Ex} = F_D \cdot \sin 30^\circ \\ \sum M_B(F) = 0: P \times 0.25 + P_1 \times 0.1 + F_{Ex} \times (1 - \frac{\sqrt{3}}{10}) - F_{Ey} \cdot (\sqrt{3} + 0.1) = 0 \end{cases}$$

数学分析

$$\therefore \begin{cases} F_{Ex} = -6.92 \text{ kN} \\ F_{Ey} = -2.48 \text{ kN} \\ F_D = 13.83 \text{ kN} \end{cases}$$

3-12

先取AD整体分析(此时C处为内力, 不分析)



$$\left\{ \begin{array}{l} F_{Ax} = 0 \\ F_{Ay} + F_{By} + F_{Dy} = 10 \times 4 \\ \sum M_C(F) = 0: \\ F_{Ay} \times 4 + F_{By} \times 2 + M = F_{Dy} \times 4 \end{array} \right.$$

再取AC段:

$$\left\{ \begin{array}{l} F_{Ax} + F_{Cx} = 0 \\ F_{Ay} + F_{By} = 9 \times 2 + F_{Cy} \\ \sum M_C(F) = 0: \\ F_{Ay} \times 4 + F_{By} \times 2 = 9 \times 2 \times 1 \end{array} \right.$$

$$\therefore \left\{ \begin{array}{l} F_{Ax} = 0 \\ F_{Cx} = 0 \end{array} \right.$$

$$\left\{ \begin{array}{l} F_{Ay} = -22.5 \text{ kN} \\ F_{By} = 55 \text{ kN} \\ F_{Cy} = 12.5 \text{ kN} \\ F_{Dy} = 7.5 \text{ kN} \end{array} \right.$$

3-22

$$\left\{ \begin{array}{l} F_{Ay} + F_{Dy} = 90 \times 2 \\ F_{Az} + F_{Dz} = 150 \times 2 \\ \Sigma M_{By}(F) = 0 : F_{Ay} \times 15 = F_{Dy} \times 30 \\ \Sigma M_{Cz}(F) = 0 : F_{Az} \times 25 = F_{Dz} \times 20 \end{array} \right.$$

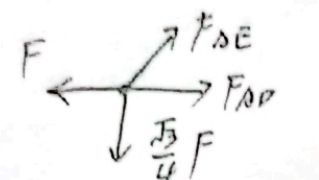
$$\therefore \left\{ \begin{array}{l} F_{Ay} = 60 \text{ N} \\ F_{Az} = \frac{400}{3} \text{ N} \\ F_{Dy} = 120 \text{ N} \\ F_{Dz} = \frac{500}{3} \text{ N} \end{array} \right.$$

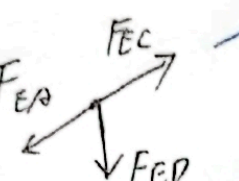
4.2 对整体受力分析求支反力

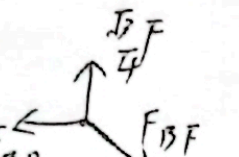
$$\begin{cases} F_{Ax} + F = 0 \\ F_{Ay} + F_{By} = 0 \\ \sum M_A(F) = 0 : F \times \frac{\sqrt{3}}{2} = F_{By} \times 2 \end{cases}$$

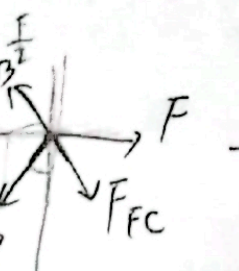
$$\therefore \begin{cases} F_{Ax} = -F \\ F_{Ay} = -\frac{\sqrt{3}}{4}F \\ F_{By} = \frac{\sqrt{3}}{4}F \end{cases}$$

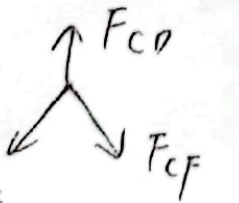
下用节点法作答:

A:  $\longrightarrow F_{AE} = \frac{F}{2}$

E:  $\longrightarrow F_{EC} = -F_{EA} = F_{AF} = \frac{F}{2}$ (在 180° 平面内)

B:  $\longrightarrow F_{BF} = \frac{F}{2}$

F:  $\longrightarrow F_{FC} = -\frac{F}{2}$

$\therefore$ C:  $\therefore F_{CD} = \frac{\sqrt{3}}{2}F$

4-3

截面法:

整体分析:

$$\begin{cases} F_{Ax} + F_{Bx} = 0 \\ F_{Ay} + F_{By} = 4F \end{cases}$$

$$\begin{cases} F_{Ax} + F_{Bx} = 0 \\ F_{Ay} + F_{By} = 4F \end{cases}$$

$$\Sigma M_B(F) = 0: F \times a + F \times 2a + F \times 3a + F \times 4a + F_{Ax} \times 2a = 0$$

$$\therefore F_{Ax} = -5F (\text{左}) \quad F_{Bx} = 5F (\text{右})$$

$$\begin{cases} F_{Ay} = \frac{5F}{2} \\ F_{By} = \frac{3F}{2} \end{cases}$$

将 1. 2. 3 杆截断, 取左半部分分析 (受力分析)

$$\begin{cases} F_{Ax} + F_{Bx} + F_1 \times \frac{2}{\sqrt{5}} + F_2 \times \frac{2}{\sqrt{13}} + F_3 = 0 \\ F_{Ay} + F_{By} - F_1 \times \frac{1}{\sqrt{5}} + F_2 \times \frac{3}{\sqrt{13}} = 0 \end{cases}$$

$$\begin{cases} F_{Ax} + F_{Bx} + F_1 \times \frac{2}{\sqrt{5}} + F_2 \times \frac{2}{\sqrt{13}} + F_3 = 0 \\ F_{Ay} + F_{By} - F_1 \times \frac{1}{\sqrt{5}} + F_2 \times \frac{3}{\sqrt{13}} = 0 \end{cases}$$

$$\Sigma M_B(F) = 0: F_{Ax} \times 2a + F_1 \times \frac{2}{\sqrt{5}} \times 2a = 0$$

$$\therefore \begin{cases} F_1 = \frac{5\sqrt{5}}{2} F \\ F_2 = -\frac{\sqrt{13}}{2} F \\ F_3 = -4F \end{cases}$$

11. 3 (山峰值高)

12. 3 (斜率大)

13. 1 (作平行线后与 ε 轴的交点大)

6-7 (1) 对于无明显屈服阶段的塑性材料, 取 $\varepsilon=0.2\%$ 处的应力作为条件屈服强度

(2) 不对, 应在 ε 轴上先找到0.2, 后过该点作斜率为弹性模量的直线, 取与曲线的交点为 $\sigma_{0.2}$

6-13 横截面 正 拉

斜截面 切 剪

6-16 (1) oa (直线段) 0.1 为弹性变形

$0.1a$ 为塑性变形

(2) 0.3 对应的数为伸长率

习题部分

6-2 设BC杆轴力为 F , 1、2件连接处支持力为 F , 记约束处分别为A、B; 隔离分析得:

$$\therefore \sum M_A = 0 : F \times 3 = F_1 \times 4.5 + F \times 1.5$$

$$\left\{ \begin{array}{l} \sum M_B = 0 : (-F_1) \times 1.5 = -F \times 0.75 \end{array} \right.$$

$$\therefore F = 6 \text{ kN}$$

$$\therefore \sigma = \frac{F}{A} = 76.4 \text{ MPa}$$

6-3 应发生在A最小处

有洞部分的面积分别为 560, 800, 840

$$\therefore \sigma_{\max} = \frac{F}{S_{\min}} = 67.9 \text{ MPa}, \text{ 在最上方的洞截面中}$$

6-7

$$F = P \cdot A = 1 \times 10^6 \times \pi \times \left(\frac{35}{2}\right)^2 \times 10^{-4} \text{ N} = 6 \cdot F_N$$

$$\therefore \sigma = \frac{F_N}{\pi \cdot \frac{d^2}{4}} \leq [\sigma] \quad \therefore d \geq 22.6 \text{ mm}$$

$\therefore$ 内径为 24 mm (取偶厚则)

6-10:

(1). 在“拉断”的前提下! $d_{\max}$.

$$\therefore \frac{F_N}{A} \geq \sigma_b$$

$$\therefore d \leq 2 \sqrt{\frac{F_{N,\max}}{\pi \cdot \sigma_b}} = 17.84 \text{ mm}$$

$$(2). \frac{F_N}{A} \leq [\sigma] = \frac{\sigma_s}{n} \quad (\text{不能超过安全范围})$$

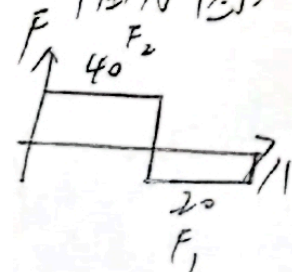
$$\therefore A \geq \frac{2 F_{N,\max}}{\sigma_s} = 833 \text{ mm}^2$$

$$(3). \sigma = \frac{F_N}{A} \leq \sigma_p \quad (\text{在弹性限度内})$$

$$\therefore F_{N,\max}' = \sigma_p \cdot A = 15.7 \text{ kN}$$

6-12:

轴力图:



$$\Delta L_1 = \frac{F_1 \cdot L}{E \cdot A_1} = -0.025 \text{ mm}$$

$$\Delta L_2 = \frac{F_2 \cdot L_2}{E \cdot A_2} = 0.1 \text{ mm}$$

$$\therefore \Delta L = \Delta L_1 + \Delta L_2 = 0.075 \text{ mm}$$

0-11 记 B、D 处钢索上的力为 F_B, F_D (沿索向上)

$$\therefore \begin{cases} F_B = F_D \\ F_{Ax} = 0 \\ F_{Ay} + 2F_B \sin 60^\circ = F \\ \sum M_A(F) = 0: F_B \times \sin 60^\circ \times 800 + F_D \times \sin 60^\circ \times 1600 = F \times 1200 \end{cases}$$

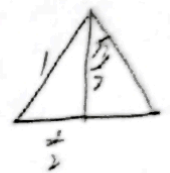
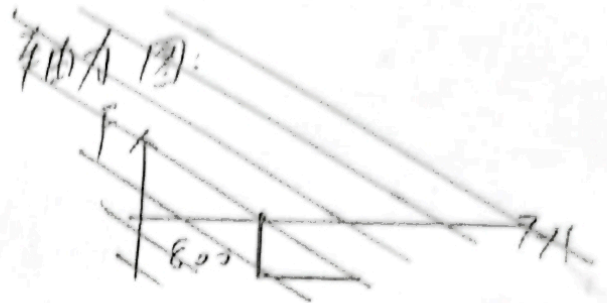
$$\therefore \begin{cases} F_{Ax} = 0 \\ F_B = 11.5 \text{ kN} \end{cases}$$

$$\therefore \sigma = \frac{F_D}{A} \approx 151 \text{ MPa}$$

$\therefore$ C 处下降可通过钢索的拉伸反应

钢索拉伸: $\Delta l = \frac{F_B \cdot l}{E \cdot A} \approx 0.68 \text{ mm}$

$$\therefore \Delta h = \frac{\Delta l}{\frac{\sqrt{3}}{2}} = 0.79 \text{ mm}$$

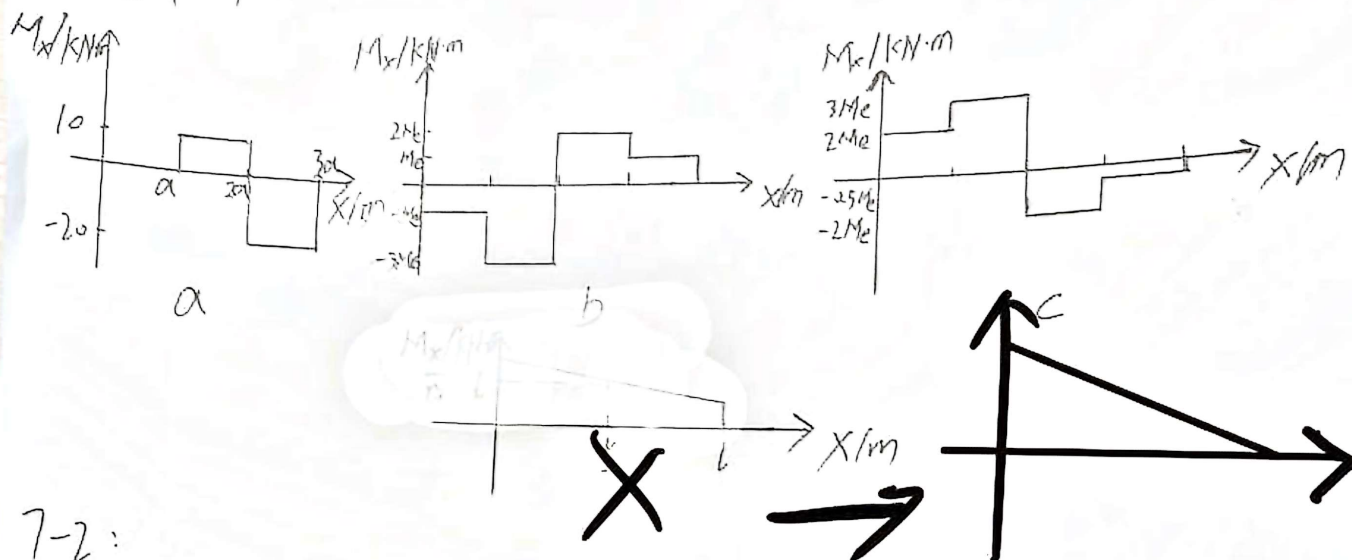


20200610

思考题:

7-7: a, c 正确

习题 7-1:



7-2:

1-1: $M_{1x} = 900 \text{ N}\cdot\text{m}$ 2-2: $M_{2x} = 400 \text{ N}\cdot\text{m}$

将 $900 \text{ N}\cdot\text{m}$ 的轮子放在中间

7-7: $M_{e1} = 9549 \times \frac{13}{200} = 621 \text{ N}\cdot\text{m}$ $M_{e2} = M_{e3} - M_{e1} = 811 \text{ N}\cdot\text{m}$

$M_{e3} = 9549 \times \frac{30}{200} = 1432 \text{ N}\cdot\text{m}$

$W_{P1} = \frac{\pi d_1^3}{16} = 4\pi \times 10^{-6} \text{ cm}^3$ $W_{P2} = \frac{\pi d_2^3}{16} = 21.4\pi \times 10^{-6} \text{ cm}^3$

$\therefore$ 强度: $\sigma_{\max 1} = \frac{M_{e1}}{W_{P1}} = 49.4 \text{ MPa}$ $\sigma_{\max 2} = \frac{M_{e3}}{W_{P2}} = 21.3 \text{ MPa}$
均小于 $[\sigma]$

刚度: $I_{P1} = \frac{\pi d_1^4}{32}$ $I_{P2} = \frac{\pi d_2^4}{32}$

$\therefore \theta_{\max 1} = \frac{M_{e1}}{G \cdot I_{P1}} \cdot \frac{180}{\pi} = 1.77^\circ/\text{m} < [\theta]$

$\theta_{\max 2} = \frac{M_{e3}}{G \cdot I_{P2}} \cdot \frac{180}{\pi} = 2.43^\circ/\text{m} < [\theta]$

$\therefore$ 均符合

7-12:

$$(11) \therefore \begin{cases} \frac{M_{\lambda \max}}{W_p} \leq [\tau] \\ M_{\lambda \max} = 9549 \cdot \frac{P}{n} \\ W_p = \frac{\pi d^3}{16} \end{cases} \quad \begin{cases} \frac{M_{\lambda \max}}{G \cdot I_p} - \frac{180}{\pi} \leq [\theta] \\ M_{\lambda \max} = 9549 \cdot \frac{P}{n} \\ I_p = \frac{\pi d^4}{32} \end{cases}$$

$$\therefore \begin{cases} d_1 \geq 80 \text{ mm} \\ d_1 \geq 84.6 \text{ mm} \end{cases}$$

$$\begin{cases} d_2 \geq 67 \text{ mm} \\ d_2 \geq 74.5 \text{ mm} \end{cases}$$

$$\therefore d_1 = 85 \text{ mm} \quad d_2 = 75 \text{ mm}$$

(2). $d = 85 \text{ mm}$

(3). 主动轮位于从动轮之间

7-29:

剪切: $\tau = \frac{F}{b \times h} = 0.952 \text{ MPa}$

挤压: $\sigma_{bs} = \frac{F}{b \times c} = 7.41 \text{ MPa}$

7-32:

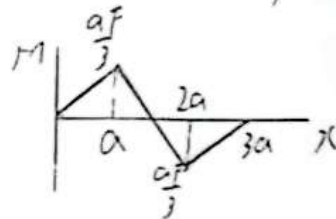
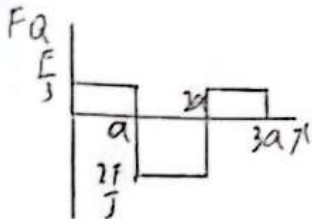
$$\begin{cases} \sigma = \frac{F}{\pi \cdot \frac{D^2}{4}} \leq 160 \text{ MPa} \rightarrow F \leq 1256 \text{ kN} \\ \tau = \frac{F}{\pi r \cdot h} \leq 100 \text{ MPa} \rightarrow F_{\text{剪}} \leq 1100 \text{ kN} \\ \sigma_{bs} = \frac{F}{\pi (R_1^2 - R_2^2)} \leq 120 \text{ MPa} \rightarrow F_{\text{挤}} \leq 7536 \text{ kN} \end{cases}$$

$$\therefore F_{\text{剪}} \leq 1100 \text{ kN}$$



8-1

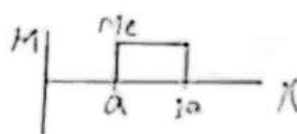
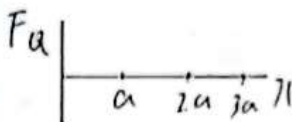
$$a): \begin{cases} \sum F_y = 0: F_{Ay} + F_{By} + F - F = 0 \\ \sum M_A = 0: F \cdot a = F \cdot 2a + F_{By} \cdot 3a \end{cases} \therefore \begin{cases} F_{Ay} = \frac{F}{3} \\ F_{By} = -\frac{F}{3} \end{cases}$$



$$\therefore |F_Q|_{\max} = \frac{2F}{3}$$

$$|M|_{\max} = \frac{aF}{3}$$

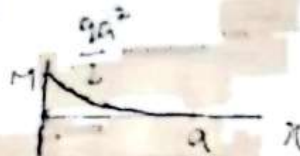
$$b): \begin{cases} \sum F_y = 0: F_{Ay} + F_{By} = 0 \\ \sum M_A = 0: -M_e + M_e + F_{By} \cdot 3a = 0 \end{cases} \therefore \begin{cases} F_{Ay} = 0 \\ F_{By} = 0 \end{cases}$$



$$|F_Q|_{\max} = 0$$

$$|M|_{\max} = M_e$$

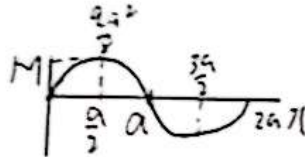
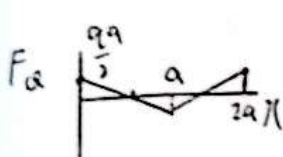
$$c): F_{Ay} = -qa$$



$$|F_Q|_{\max} = qa$$

$$|M|_{\max} = \frac{1}{2} qa^2$$

$$d): \begin{cases} \sum F_y = 0: F_{Ay} + F_{By} - qa + qa = 0 \\ \sum M_A = 0: F_{By} \cdot 2a + qa \cdot \frac{3a}{2} - qa \cdot \frac{a}{2} = 0 \end{cases} \therefore \begin{cases} F_{Ay} = \frac{qa}{2} \\ F_{By} = -\frac{qa}{2} \end{cases}$$

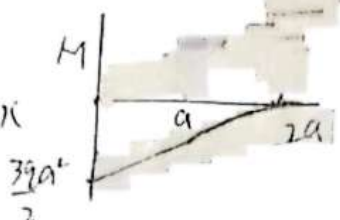
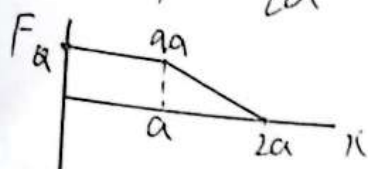


$$|F_Q|_{\max} = \frac{qa}{2}$$

$$|M|_{\max} = \frac{qa^2}{8}$$

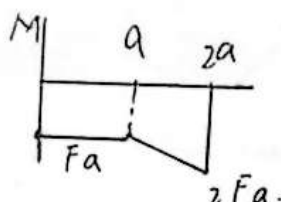
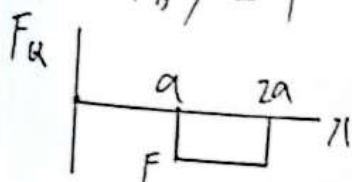


e): $F_{Ay} = qa$



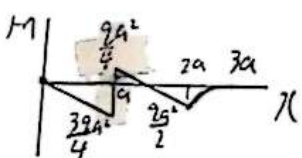
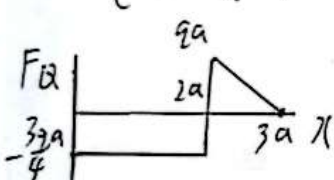
$\therefore |F_u|_{\max} = qa$
 $|M|_{\max} = \frac{3qa^2}{2}$

f): $F_{By} = F$



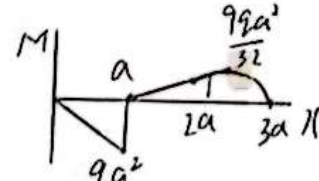
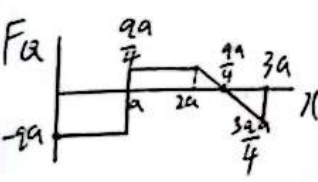
$\therefore |F_u|_{\max} = F$
 $|M|_{\max} = 2Fa$

g): $\begin{cases} \sum F_y = 0: F_{Ay} + F_{By} = qa \\ \sum M_A = 0: -qa^2 + F_{By} \cdot 2a - qa \cdot \frac{5a}{2} = 0 \end{cases} \therefore \begin{cases} F_{Ay} = -\frac{3qa}{4} \\ F_{By} = \frac{7qa}{4} \end{cases}$

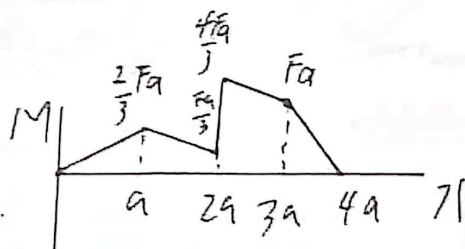
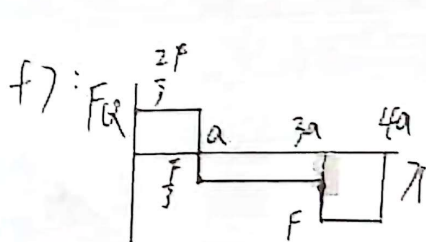
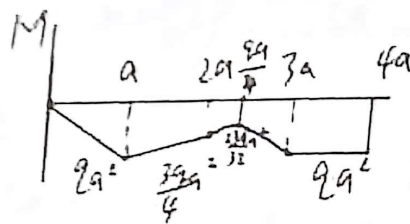
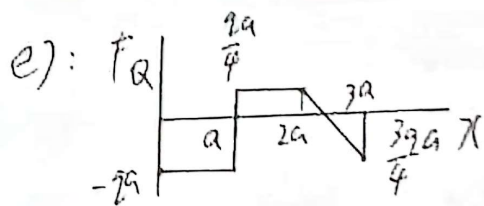
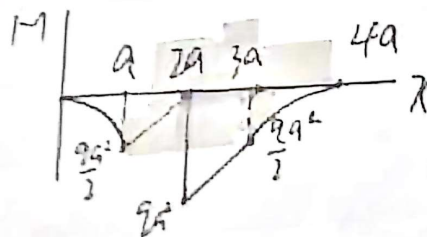
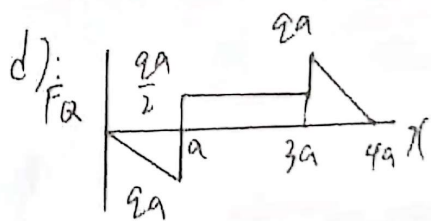


$\therefore |F_u|_{\max} = qa$
 $|M|_{\max} = \frac{3}{4}qa^2$

h): $\begin{cases} \sum F_y = 0: F_{Cy} + F_{By} = qa + qa \\ \sum M_B = 0: qa \cdot 3a - qa^2 + qa \cdot \frac{a}{2} - F_{Cy} \cdot 2a = 0 \end{cases} \therefore \begin{cases} F_{Cy} = \frac{5qa}{4} \\ F_{By} = \frac{3qa}{4} \end{cases}$



$\therefore |F_u|_{\max} = qa$
 $|M|_{\max} = qa^2$





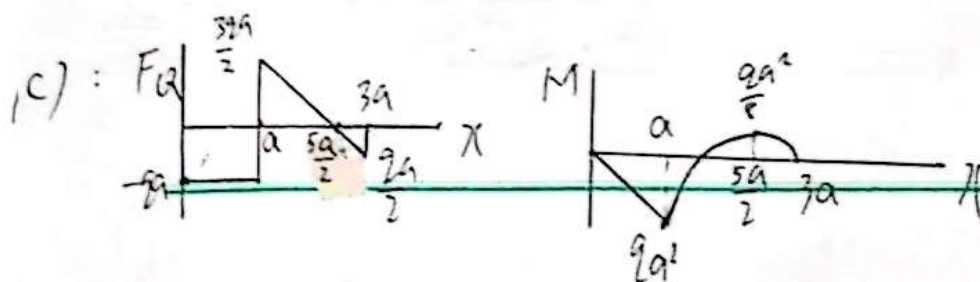
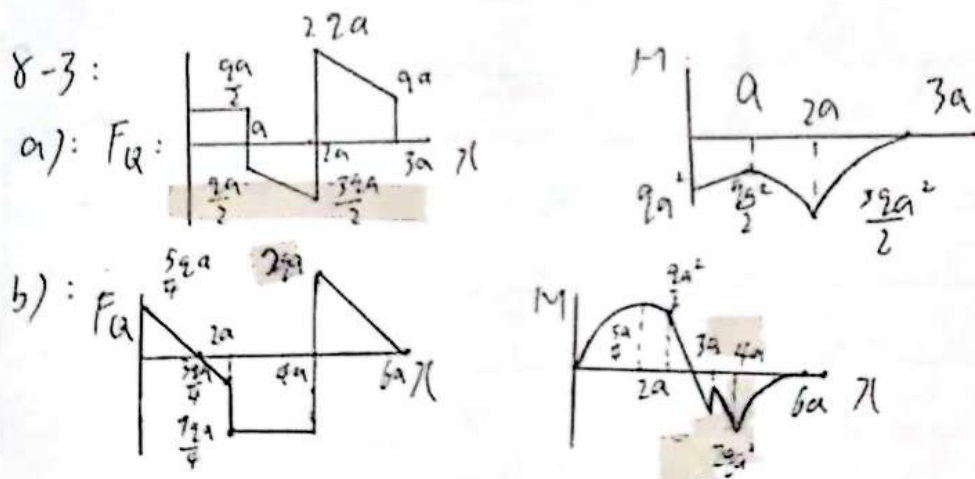
2): $\begin{cases} \sum M_C = 0: F_{Ay} + F_{Cy} - 2a \\ \sum M_A = 0: 9a \cdot \frac{3a}{2} - F_{Cy} \cdot 2a = 0 \end{cases} \therefore \begin{cases} F_{Ay} = \frac{9a}{4} \\ F_{Cy} = \frac{59a}{4} \end{cases}$

~~错, 看下一页~~

$|F_Q|_{\max} = 9a$
 $|M|_{\max} = 9a^2$

1): $\begin{cases} \sum M_C = 0: \\ \sum M_B = 0: \end{cases} \rightarrow \begin{cases} F_{Cy} = \frac{5}{2}9a \\ F_{By} = -\frac{9a}{2} \end{cases}$

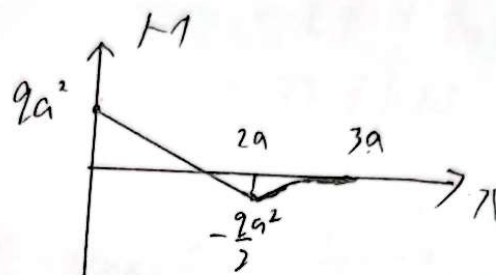
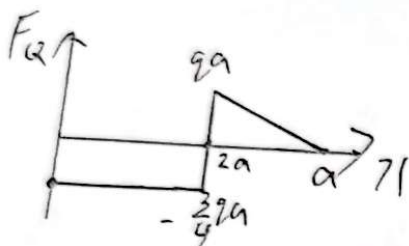
$|F_Q|_{\max} = \frac{39a}{2}$
 $|M|_{\max} = 9a^2$



8-1 z

$$\begin{cases} F_{Ay} + F_{Cy} = qa \\ qa^2 = F_{Cy} \times 2a - qa \times \frac{5a}{2} \end{cases} \therefore \begin{cases} F_{Ay} = -\frac{3}{4}qa \\ F_{Cy} = \frac{7}{4}qa \end{cases}$$

剪力图:



9-8

解:

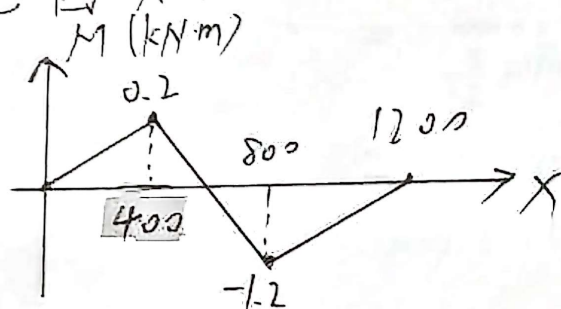
$$\therefore \sum M(A) = 0$$

$$\therefore F_B = 0.5 \text{ kN}$$

$$\therefore \sum M(B) = 0$$

$$\therefore F_A = 0.5 \text{ kN}$$

$\therefore$ 弯矩图如下:



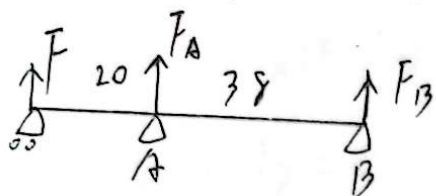
$$\therefore \sigma = \frac{M \cdot y}{I_z}$$

$$\therefore \sigma_1 = \frac{0.2 \times 30 \times 10^{-1}}{\frac{\pi}{64} (60 \times 10^{-4})^4} = 9.9 \times 10^9 \text{ Pa}$$

$$\sigma_2 = \frac{1.2 \times 30 \times 10^{-1}}{\frac{\pi}{64} [(60 \times 10^{-4})^4 - (45 \times 10^{-4})^4]} = 8.29 \times 10^{10} \text{ Pa}$$

$$\therefore \sigma_{\max} = 8.29 \times 10^{10} \text{ Pa}$$

9-11 原图可抽象为:



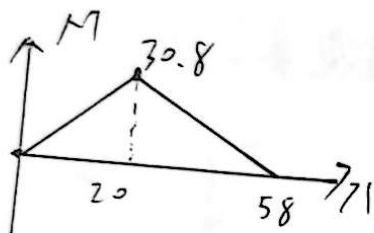
$$\therefore \sum M_A(F) = 0$$

$$\therefore 20 \times F = F_B \times 38$$

$$\therefore F_B = 8.1 \text{ kN}$$

$$\therefore F_A = -(F + F_B) = -23.5 \text{ kN}$$

$\therefore$ 弯矩图:



$$\begin{aligned} \therefore \sigma_{\max} &= \frac{M_{\max} \cdot y}{I_z} \\ &= \frac{30.8 \times 10 \times 10^{-4}}{\frac{1}{12} \times 30 \times (20^3 - 12^3) \times 10^{-6}} = 197.1 \text{ MPa} \end{aligned}$$

$$\therefore [\sigma] = \frac{\sigma_s}{n} = 253.3 \text{ MPa}$$

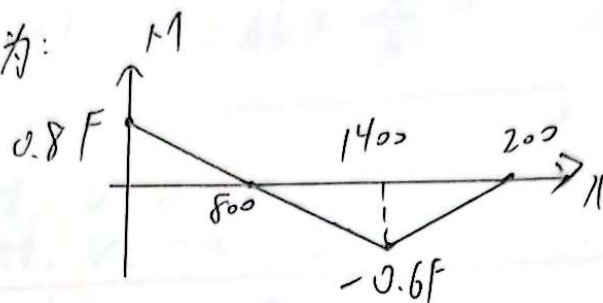
$$\therefore \sigma_{\max} \leq [\sigma]$$

$\therefore$ 安全

9-13

$$\begin{cases} \sum M_A(F) = 0 \\ \sum F_y = 0 \end{cases} \therefore \begin{cases} F_{Ay} = -F \\ M_A = 0.8F \end{cases}$$

$\therefore$ 弯矩图 为:



$\therefore$ C 处负弯矩最大, A 处正弯矩最大

C 处:



$\therefore$ 上半部分受压, 下半部分受拉

$$\therefore \sigma = \frac{Mh}{I}$$

$$\therefore \begin{cases} \sigma_{压} = \frac{0.6F \times h_2}{I_z} \leq [\sigma_c] \\ \sigma_{拉} = \frac{0.6F \times h_1}{I_z} \leq [\sigma_t] \end{cases}$$

A 处:



上受拉, 下受压

$$\therefore \begin{cases} \sigma_{压} = \frac{0.8F \times h_1}{I_z} \leq [\sigma_c] \\ \sigma_{拉} = \frac{0.8F \times h_2}{I_z} \leq [\sigma_t] \end{cases}$$

$\therefore$ 解得 $F \leq 44H$

$$\therefore [F] = 44N$$

10-1

a: $\pi=0$ 时, $V=0$
 $\pi=\frac{l}{2}$ 时, $V_c=V_{ct}$, $\theta_c=\theta_{ct}$
 $\pi=l$ 时, $V=\Delta l = \frac{F}{k}$

b: $\pi=0$ 时, $V=0$, $\theta=0$
 $\pi=\frac{l}{2}$ 时, $V_c=V_{ct}$, $\theta_c=\theta_{ct}$

c: $\pi=0$ 时, $V=0$, $\theta=0$
 $\pi=a$ 时, $V_c=V_{ct}$
 $\pi=a+l$ 时, $V=0$

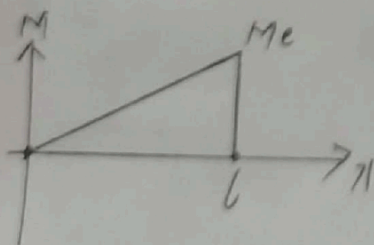
d: $\pi=0$ 时, $V=0$,
 $\pi=2a$ 时, $V=\Delta l = \frac{F_A l}{EA}$
 $\pi=a$ 时, $V_c=V_{ct}$, $\theta_c=\theta_{ct}$

10-1

$\sum M_A(F) = 0$
 $F_{Ay} + F_{By} = 0$

$\rightarrow \begin{cases} F_{Ay} = \frac{Me}{l} \\ F_{By} = -\frac{Me}{l} \end{cases} \quad \begin{aligned} &\therefore f(A) = 0 \\ &\therefore D = 0 \\ &\therefore f = \frac{F_A}{EI} \cdot \frac{\pi^3}{6} + C\pi \end{aligned}$

弯矩图如图:



$\therefore M(x) = \frac{Me}{l} \cdot x$

$(0 \leq x < l)$

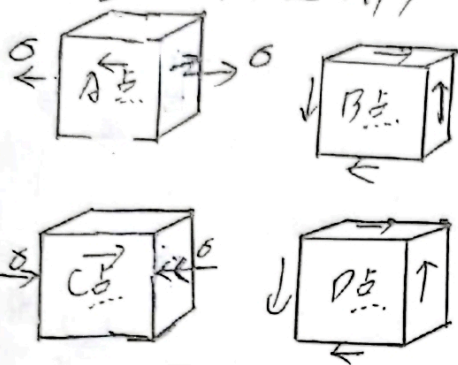
$\therefore \theta = \int_0^l \frac{M(x)}{EI} dx \quad \therefore f(B) = 0$
 $= \frac{F_A}{EI} \cdot \frac{\pi^2}{2} + C \quad \therefore C = -\frac{F_A l^2}{6EI} = -\frac{Me l}{6EI}$
 $\therefore \theta_A = -\frac{Me l}{6EI}$
 $\therefore f = \int_0^l \theta dx$
 $= \frac{F_A}{EI} \cdot \frac{\pi^3}{6} + C\pi + D \quad f_{\frac{l}{2}} = -\frac{Me l^2}{16EI}$

第十一章:

11-4: F 向中心处移动

得弯曲力: $F = F$

扭矩: M (逆时针)



$$\begin{cases} \sigma_{max} = \frac{\sigma}{2} + \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \\ \sigma_x = 0 \\ \sigma_{min} = \frac{\sigma}{2} - \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2} \end{cases}$$

$$\therefore \sigma_1 = 56.1 \text{ MPa}$$

$$\sigma_2 = 0$$

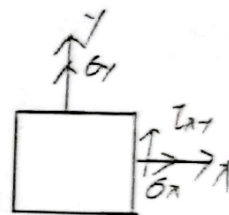
$$\sigma_3 = -16.1 \text{ MPa}$$

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2} = 36.1 \text{ MPa}$$

11-10: $\sigma_x = 0$ $\sigma_y = 0$

$$\sigma_x = \frac{F}{A} = \frac{20 \times 10^3}{\pi \times 4 \times 10^{-4}} = 15.92 \text{ MPa}$$

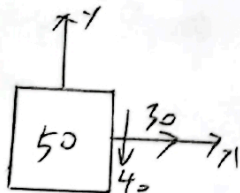
$$\tau_{xy} = -\frac{M_e}{W_p} = -\frac{0.8 \times 10^3}{\frac{\pi}{16} \times 64 \times 10^6} = -63.66 \text{ MPa}$$



$$\begin{cases} \sigma_x = 0 \\ \sigma_{max} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = 72.1 \text{ MPa} \\ \sigma_{min} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = -56.3 \text{ MPa} \end{cases}$$

$$\therefore \sigma_1 = 72.1 \text{ MPa}, \sigma_2 = 0, \sigma_3 = -56.3 \text{ MPa}$$

11-12 C:

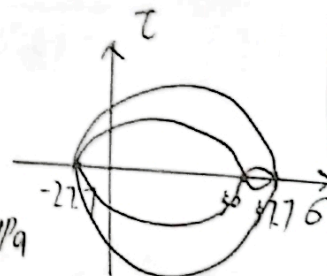


$$\begin{cases} \sigma_x = 50 \text{ MPa} \\ \sigma_y = 30 \text{ MPa} \\ \sigma_z = 0 \\ \tau_{xy} = 40 \text{ MPa} \end{cases}$$

$$\begin{cases} \sigma_{max} = 15 + 42.7 = 57.7 \text{ MPa} \\ \sigma_{min} = 15 - 42.7 = -27.7 \text{ MPa} \end{cases}$$

$$\therefore \sigma_1 = 57.7 \text{ MPa}, \sigma_2 = 50 \text{ MPa}, \sigma_3 = -27.7 \text{ MPa}$$

$$\tau_{max} = \frac{\sigma_{max} - \sigma_{min}}{2} = 42.7 \text{ MPa}$$



第十二章:

12-5: (拉伸与弯曲)

$$\cos \varphi = \frac{25}{26.2} \approx 0.95$$

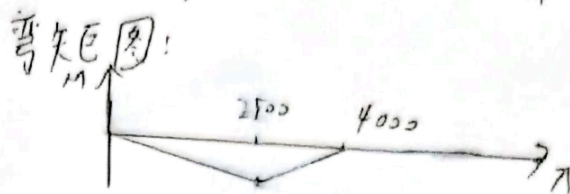
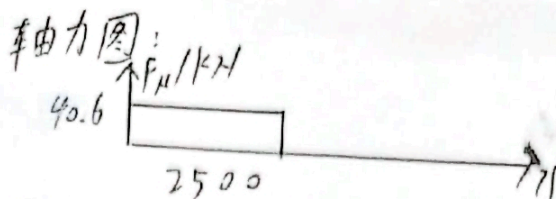
$$\sin \varphi = \frac{8}{26.2} \approx 0.3$$

$$\therefore \begin{cases} F_{Ax} = F \cdot \cos \varphi \\ F_{Ay} + F \cdot \sin \varphi = G \\ \sum M(A) = 0 : F \cdot \sin \varphi \cdot 2500 = G \cdot 4000 \end{cases}$$

$$\therefore \begin{cases} F = 42.7 \text{ kN} \\ F_{Ax} = 40.6 \text{ kN} \\ F_{Ay} = -4.8 \text{ kN} \end{cases}$$

F_{Ax} , $F \cos \varphi$ 产生拉伸变形

F_{Ay} , $F \sin \varphi$, G 产生弯曲变形



$$\therefore \sigma_{max} = \left| \frac{F_N}{A} \right| + \left| \frac{9.8 \times 2.5}{W_z} \right| \leq [\sigma] = 100 \text{ MPa}$$

选 16 号钢

$$A \approx 26.13 \text{ cm}^2$$

$$W_z = 141 \text{ cm}^3$$

$$\sigma_{max} \approx 100.6$$

12-10: (弯曲与扭转)

将两力向中心平行, 得

$$F_z = F + 3F = 4F$$

$$M_{e1} = (3F - F) \cdot \frac{D}{2} = F \cdot D$$

$$\therefore M_{e2} = 9549 \cdot \frac{P}{n}$$

$$= 1021.7 \text{ N}\cdot\text{m}$$

$\therefore$ 输入等于输出

$$\therefore F \cdot D = 1021.7 \text{ N}\cdot\text{m}$$

$$\therefore F = 1.55 \text{ kN}$$

$\therefore$ 危险点为 A 点

$$\sigma_{max} = \frac{4 \times F \times l}{\frac{\pi d^3}{32}} \quad (\text{弯曲})$$

$$\tau_{max} = \frac{M_{e2}}{\frac{\pi d^3}{16}}$$

$$\therefore \begin{matrix} \sigma_{max} \\ \sigma_{min} \end{matrix} = \frac{\sigma}{2} \pm \sqrt{\left(\frac{\sigma}{2}\right)^2 + \tau^2}$$

$\therefore$ 根据第三强度理论

$$\sigma_3 = \sigma_1 - \sigma_3 = \sqrt{\sigma^2 + 4\tau^2} \leq [\sigma] = 70 \text{ MPa}$$

$$\therefore [l] = 2.54 \text{ m}$$

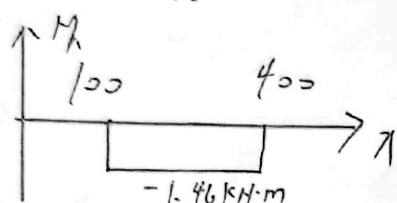
12-16

$$M_e = 9549 \cdot \frac{P}{n} = 1.46 \text{ kN}\cdot\text{m}$$

$$\therefore F_z = \frac{M_e}{\frac{1}{2}\pi r^2} = 5.84 \text{ kN}$$

$$F_y = \frac{M_e}{\frac{1}{2}\pi r^2} = 7.3 \text{ kN}$$

扭矩图如图:

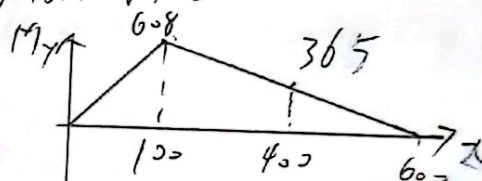


$$\left\{ \begin{aligned} \sum M_{Ay}(F) = 0: 100 \times F_y &= F_{By} \times 600 \\ F_{Ay} + F_{By} &= F_y \end{aligned} \right.$$

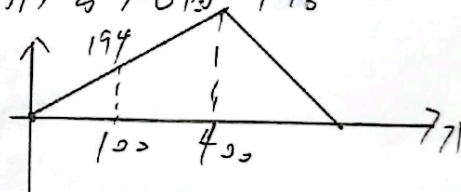
$$\therefore \begin{cases} F_{Ay} = 6.08 \text{ kN} \\ F_{By} = 1.22 \text{ kN} \end{cases}$$

$$\text{同理: } \begin{cases} F_{Az} = 1.94 \text{ kN} \\ F_{Bz} = 3.9 \text{ kN} \end{cases}$$

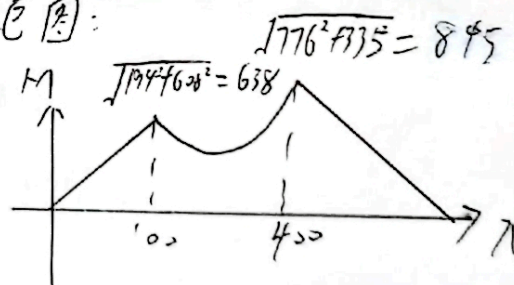
y方向弯矩图:



z方向弯矩图:



合成弯矩图:



$$\therefore M_{\max} = 845 \text{ N}\cdot\text{m}$$

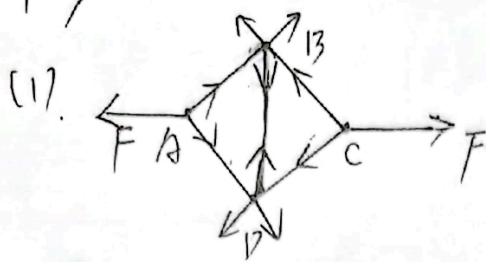
$$M_{T, \max} = 1460 \text{ N}\cdot\text{m}$$

$\therefore$ 利用第三强度理论

$$\sigma_{r3} = \frac{1}{W} \sqrt{M_{\max}^2 + M_{T, \max}^2} \leq [\sigma] = 60 \text{ MPa}$$

$$\therefore d \geq 65.8 \times 10^{-3} \text{ m} \quad \therefore d = 66 \text{ mm}$$

14-3



BD 为压杆 $\lambda = \frac{\mu l}{z}, z = \frac{d}{4}, \mu = 1$

↓
压力为 F $\therefore \lambda = 113 > \lambda_p = 100$

$\therefore$ 为大柔度杆

$\therefore$ 最大载荷

$\therefore F_{cr} = \sigma_{cr} \cdot A = \frac{\pi^2 E}{\lambda^2} \cdot A = 303 \text{ kN}$ 为 303 kN

(2) F 反向后, BD 为拉杆, 周围四杆为压杆

$\therefore$ BD: $F = \sigma_s \cdot A = 471 \text{ kN}$

↓
压力为 $\frac{F}{\sqrt{2}}$

AB: $\lambda = \frac{\mu l}{z} = 80$

$60 = \lambda_s < 80 < \lambda_p = 100$

$\therefore$ AB 为中柔度杆 $\begin{matrix} 304 \\ \nearrow \\ 1.12 \end{matrix}$

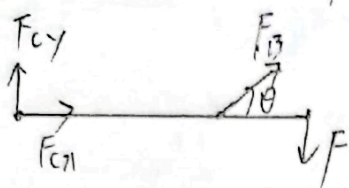
$\therefore F = \sqrt{2} \times \sigma_{cr}' \times A = \sqrt{2} (a - b\lambda) \cdot A = 595 \text{ kN}$

$\therefore$ 临界载荷为 471 kN

$\therefore$ 最大载荷为 471 kN

14-5

11. AB 为压杆 (根据 C 处矩为 0 可得)



$$\therefore \lambda = \frac{\mu l}{i}, \quad i = \frac{d}{4}$$

$$\therefore \lambda = 80$$

$\therefore$ 为中柔度杆

$$\therefore F_{cr}' = \sigma_{cr} \cdot A = (a - b\lambda) \cdot \frac{\pi d^2}{4} = 269.51 \text{ kN}$$

$$\sin \theta = \frac{\sqrt{3}}{4}$$

$$\therefore (F_{cr}' \cdot \sin \theta) \times 600 = F_{cr} \times 900$$

$$\therefore F_{cr} = 118 \text{ kN}$$

12.

$$\therefore [F] = \frac{118}{n_{st}} = \frac{118}{2} = 59 \text{ kN} < 60 \text{ kN}$$

$\therefore$ 不稳定

14-17

先检查强度条件(AB是否会断)

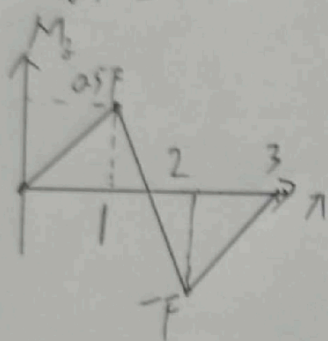
对A点取矩:

$$2F \times 1 + F \times 3 = F_c \times 2$$

$$\therefore F_c = 2.5F \quad (\text{CD杆受压})$$

$$\therefore F_A = 2F + F - F_c = 0.5F$$

其弯矩图为:



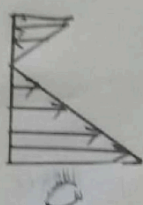
$\therefore$ 危险截面为C截面

$\therefore$ 为铸铁

$\therefore$ 抗拉强度与抗压强度不同

上半部分抗压

下半部分抗拉



$$W_1 = \frac{I_z}{Y_1} = 1.6 \times 10^{-4} \quad W_2 = \frac{I_z}{Y_2} = 8.89 \times 10^{-5}$$

$$\therefore \sigma_{cc} = \frac{M_c}{W_2} = \frac{F}{W_2} \leq 120 \text{ MPa}$$

$$\therefore F \leq 6.4 \text{ kN}$$

$$\sigma_{ct} = \frac{M_c}{W_1} = \frac{F}{W_1} \leq 40 \text{ MPa}$$

再检查失稳条件

$$\lambda = \frac{\mu L}{i} = 115.57 \lambda_p$$

$\therefore$ CD为大柔度杆

$$i = \sqrt{\frac{I}{A}} = \sqrt{\frac{bh^3}{12}} = 8.66 \times 10^{-3} \text{ m}$$

$\therefore$ 许可载荷为6.4 kN

$$\therefore F_{cr} = G_{cr} \cdot A = \frac{\pi^2 E}{\lambda^2} A = 222 \text{ kN}$$

$$\therefore [F] = \frac{F_{cr}}{n_{st}} = 74 \text{ kN} > 6.4 \text{ kN}$$